

2019

COMPUTER SCIENCE

Paper : CSM-101

(Advances in Computer Architecture)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*1. Answer **any five** questions :

2×5

- (a) Define 'instruction execution'.
- (b) Define 'systolic architecture'.
- (c) Differentiate RISC from CISC.
- (d) Derive the formula for average access time in a three-level cache memory architecture.
- (e) How many states are there in 4×4 switch including broadcasting and permutation?
- (f) State the effect of clock-skewing.
- (g) Explain why decoding of instructions in VLIW is efficient than super-scalar or other parallel architecture.

2. Answer **any five** questions :

4×5

- (a) Estimate the effect of branch instructions on the performance of pipelined architectures with suitable set of parameters.
- (b) For a given latency cycle, split out the design steps to get a reservation table. Explain with an example with latency cycle (2,3,2,5).
- (c) State and explain the necessary condition for a MICN to be non-blocking one. Comment on the design alternatives to convert a blocking MICN to its equivalent non-blocking MICN.
- (d) What are the basic assumptions of any message passing system? Comment on the possible states of any message passing system.
- (e) Critically comment on the desirable features of a processor, member of any multiprocessor system. What are the impacts of vector length of vector processor on vectorising compilers?
- (f) Write an algorithm to identify the conflicting pairs of source-destinations for a shuffle-exchange MICN of a given size.
- (g) A computer has a 256 KByte, 4-way set associative, write back data cache with block size of 32 Bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit. Show the cache address format for the computer system.
- (h) What is superscalar architecture? State the advantages and disadvantages of this structure.

Please Turn Over

3. Answer *any four* questions :

- (a) (i) Corresponding to the following reservation table identify the collision matrices. Mention the ORing principle to get the modified collision matrices and justify.

	0	1	2	3	4
S1	A	B		A	B
S2		A		B	
S3	B		AB		A

- (ii) For a given reservation table corresponding to pipelined architecture, what will be the upper bound of Minimal Average Latency (MAL) of incoming jobs? Justify your answer. 5+5
- (b) (i) Differentiate between Control-flow and Data-flow architectures. What are the alternative forms of Dataflow architectures? Comment on the deficiencies of DFA.
- (ii) Draw the dataflow graphs to represent the following computation :
- ```

for i ← 1 until m do
 begin
 c(i) ← 0
 for j ← 1 until n do
 c(i) ← a(i,j) x b(j)
 end
 end

```
- (iii) What is an instruction template? Give example. 4+4+2
- (c) (i) Modify the uni-processor matrix multiplication algorithm for two two-dimensional matrices to incorporate temporal and spatial localities. Suggest the suitable systolic array of processors with scheduling for the said problem. Clearly state your assumptions, if any.
- (ii) How systolic array of processors is different from SIMD environment? Explain briefly. 7+3
- (d) (i) Write algorithms to add a set of 8 integer values by multiprocessor environments where processors are in cube-connected and shuffle-exchange connected structures. Compare their time complexities.
- (ii) Compare and contrast the XOR and destination tag based routing schemes for multistage interconnection networks. 6+4
- (e) (i) Briefly discuss the alternative flow control strategies in any message passing system.
- (ii) Construct an example showing that the loss of a message can cause a deadlock among a set of communication processes.
- (iii) What is victim cache? What is a snooping cache? Discuss with example the Write Through and Write Once protocols for Cache consistency. 3+3+4

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- (f) (i) Derive an  $O(n \log_2 n)$  algorithm for SIMD matrix multiplication in a 4-cube network. Clearly show the distribution of rows of matrix A and B to different nodes of the 4-cube network.
- (ii) State the difference between vector processor and scalar processor. Explain vectorization process with respect to the following program fragment :

```
for(i=0;i<100;i++)
{
 A[i]=B[i]*C[i];
 D[i]=f(A[i]*X[i]);
}
```

4+2+4

- (g) State the difference between shared memory multiprocessor and distributed memory multiprocessor. Draw and explain how UMA, NUMA and COMA shared memory systems are working, also mention its advantages and disadvantages. 2+6+2
- (h) Assume that five stages are used to execute any instruction by a computer system and corresponding propagation time delays are shown as follows :

| Fetch | Decode | Execute | Memory | Writeback |
|-------|--------|---------|--------|-----------|
| 200ps | 150ps  | 100ps   | 190ps  | 140ps     |

Also, assume that when pipeline is used then each stage costs 20ps extra for the buffering between the stages.

Now calculate— (i) cycle time, (ii) instruction latency and (iii) throughput for the system without and with pipeline. If you could split one of the pipeline stage into two equal halves then which one will be selected— justify. Define MAL in the context of pipeline system. 3+3+2+2